Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

If $\int \frac{\sqrt{\cot x}}{\sin x \cos x} dx = a\sqrt{\cot x} + b$ find the vale of a

(A) 1

(B) 2

(C) -1

(D) -2

Q.2

$\int \left[\frac{1}{\log x} - \frac{1}{(\log x)^2}\right] dx =$

(A) $\frac{x}{\log x} + c$

(B) $x \log x + c$

(C) $\frac{\log x}{x} + c$

(D) none of these

Q.3

$\int \frac{1+x^5}{1+x} dx$ is equal to

(A) $1 - x + x^2 - x^3 + x^4 + c$

(B) $x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \frac{x^5}{5} + c$

(C) $(1 + x)^5 + c$

(D) none of these

Q.4

$\int 3^x \cdot e^x dx$ is equal to

(A) $\frac{(3e)^x}{\log(3-e)} + c$

(B) $(3e)^x + c$

(C) $\frac{(3e)^x}{1+\log 3} + c$

(D) none of these

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Q.5

$\int \frac{\cos 5x+\cos 4x}{1-2\cos 3x}dx$ is equal to

(A) $-\sin 2x-\sin x+c$

(B) $-\frac{1}{2}\sin 2x+\sin x+c$

(C) $-\frac{1}{2}\sin 2x-\sin x+c$

(D) none of these

Q.6

The value of the integral $\int \left(\frac{\ln x-1}{(\ln x)^2+1}\right)^2 dx$ is

(A) $\frac{x}{x^2+1}+c$

(B) $\frac{\ln x}{(\ln x)^2+1}$

(C) $\frac{x}{(\ln x)^2+1}+c$

(D) $e^x\left(\frac{x}{x^2+1}\right)+c$

Q.7

If $\int f(x)\cos x dx=-\frac{1}{2}f^2(x)+c$

(A) x

(B) 1

(C) $\cos x$

(D) $\sin x$

Q.8

The value of $\int \frac{d(\sin x)}{\sqrt{1-\sin^2 x}}$ is equal to

(A) $x+c$

(B) $3x+c$

(C) x^2+c

(D) none of these

Q.9

$\int e^x\frac{(1+\sin x)}{(1+\cos x)}dx$ is equal to

(A) $e^x\cos\left(\frac{x}{2}\right)+c$

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(B) $e^x \tan\left(\frac{x}{2}\right) + c$

(C) $e^x \tan x + c$

(D) $e^x \cot x + c$

Q.10

$\int \tan^{-1}\sqrt{x}dx$ is equal to

(A) $(x+1)\tan^{-1}\sqrt{x} - \sqrt{x} + C$

(B) $x \cdot \tan^{-1}\sqrt{x} - \sqrt{x} + C$

(C) $\sqrt{x} - x\tan^{-1}\sqrt{x} + C$

(D) $\sqrt{x} - (x+1)\tan^{-1}\sqrt{x} + C$

Q.11

$\int \frac{\{f(x)\cdot\phi'(x) - f'(x)\cdot\phi(x)\}}{f(x)\cdot\phi(x)}\{\log\phi(x) - \log f(x)\}dx$

(A) $\log\frac{\phi(x)}{f(x)} + k$

(B) $\frac{1}{2}\left\{\log\frac{\phi(x)}{f(x)}\right\}^2 + k$

(C) $\frac{\phi(x)}{f(x)}\log\frac{\phi(x)}{f(x)} + k$

(D) none of these

Q.12

If $\int f(x)dx = F(x)$, then $\int x^3 f\left(x^2\right)dx$ is equal to

(A) $\frac{1}{2}\left(x^2F\left(x^2\right) - \int F\left(x^2\right)dx^2\right)$

(B) $\frac{1}{2}\left(x^2F\left(x^2\right) - \int F\left(x^2\right)dx\right)$

(C) $\frac{1}{2}\left(x^2F(x) - \frac{1}{2}\int F\left(x^2\right)dx\right)$

(D) none of these

Q.13

The value of the integral $\int \frac{(1-\cos\theta)^{2/7}}{(1+\cos\theta)^{9/7}}d\theta$ is

(A) $\frac{7}{11}\left(\tan\frac{\theta}{2}\right)^{\frac{11}{7}} + C$

(B) $\frac{7}{11}\left(\cos\frac{\theta}{2}\right)^{\frac{11}{7}} + C$

(C) $\frac{7}{11}\left(\sin\frac{\theta}{2}\right)^{\frac{11}{7}} + C$

(D) None of these

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Q.14

If $\int \frac{dx}{1+e^x} = x + \mathrm{K}\log(1+e^x) + C$ then K is equal to

(A) 1

(B) -1

(C) 2

(D) -2

Q.15

If $\int \frac{2x-\sqrt{\sin^{-1}x}}{\sqrt{1-x^2}}dx = C - 2\sqrt{1-x^2} - \frac{2}{3}\sqrt{f(x)}$ then $f(x)$ is equal to

(A) $\sin^{-1}x$

(B) $2\sin^{-1}x$

(C) $\left(\sin^{-1}x\right)^3$

(D) $3\left(\sin^{-1}x\right)^3$

Q.16

If $f\left(\frac{3x-4}{3x+4}\right) = x+2$, then $\int f(x)dx$ is:

(A) $e^{x+2}\log\left|\frac{3x-4}{3x+4}\right| + c$

(B) $-\frac{8}{3}\log|1-x| + \frac{2}{3}x + c$

(C) $\frac{8}{3}\log|1-x| + \frac{x}{3} + c$

(D) $e^{[(3x-4)(3x+4)]} - \frac{x^2}{2} - 2x + c$

Q.17

If $f(x) = \frac{\log x}{x^3}$, then its antiderivative $F(x)$ given by

(A) $C - \frac{1}{2x^2}\log(x\sqrt{e})$

(B) $C + \frac{1}{2x^2}\log x$

(C) $C - \frac{1}{2x^2}\log(xe)$

(D) $C + \frac{1}{2x^3}\log(xe)$

Q.18

The value of $\int \frac{\sin x}{\sin 4x}\,\mathrm{d}x$ is

(A) $\frac{1}{4}\log\left|\frac{\sin x-1}{\sin x+1}\right| - \frac{1}{\sqrt{2}}\log\left|\frac{\sqrt{2}\sin x-1}{\sqrt{2}\sin x+1}\right| + C$

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(B) $\frac{1}{8}\log\left|\frac{\cos x-1}{\cos x+1}\right|-\frac{1}{2\sqrt{2}}\log\left|\frac{\sqrt{2}\cos x-1}{\sqrt{2}\cos x+1}\right|+C$

(C) $\frac{1}{8}\log\left|\frac{\sin x-1}{\sin x+1}\right|-\frac{1}{4\sqrt{2}}\log\left|\frac{\sqrt{2}\sin x-1}{\sqrt{2}\sin x+1}\right|+C$

(D) none of these

Q.19

Let $I=\int\frac{e^x}{e^{4x}+e^{2x}+1}dx, J=\int\frac{e^{-x}}{e^{-4x}+e^{-2x}+1}dx$. Then for an arbitrary constant C, the value of $I-J$ equals

(A) $\frac{1}{2}\log\left(\frac{e^{4x}-e^{2x}+1}{e^{4x}+e^{2x}+1}\right)+C$

(B) $\frac{1}{2}\log\left(\frac{e^{2x}+e^{x}+1}{e^{2x}-e^{x}+1}\right)+\mathrm{C}$

(C) $\frac{1}{2}\log\left(\frac{e^{2x}-e^{x}+1}{e^{2x}+e^{x}+1}\right)+C$

(D) $\frac{1}{2}\log\left(\frac{e^{4x}+e^{2x}+1}{e^{4x}-e^{2x}+1}\right)+C$

Q.20

$\int e^{\tan x}(\sin x-\sec x)dx$, is equal to

(A) $e^{\tan x}\cos x+C$

(B) $e^{\tan x}\sin x+C$

(C) $-e^{\tan x}\cos x+C$

(D) $e^{\tan x}\sec x+C$

Q.21

The $\int\frac{\sin^2 x}{\cos^6 x}\,\mathrm{d}x$ is a is a polynomial of degree n in tanx, then n =

Q.22

Let $f(x)=1+3^x\log 3$ and $F(x)$ be its antiderivative which assume the value 7 for $x=2$. The value of x for which the curve $F(x)$ cuts the abscissa axis is

Q.23

If it is known that at the point $x=1$ two antiderivative of $f(x)=e^x$ differ by 2. The difference of these antiderivatives at $x=100$ is

Q.24

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If the integral $\int \frac{5\tan x}{\tan x-2}dx = x + a\log|\sin x - 2\cos x| + k$, then a is equal to

Q.25

Let $f(x) = \int x^{\sin x}(1 + x\cos x \cdot \ln x + \sin x)dx$ and $f\left(\frac{\pi}{2}\right) = \frac{\pi^2}{4}$, then the value of $|\cos(f(\pi))|$ is

Q.26

If $\int \frac{f(x)}{x^2-x+1}\,dx = \frac{3}{2}\log\left(x^2 - x + 1\right) + \frac{1}{\sqrt{3}}\tan^{-1}\frac{2x-1}{\sqrt{3}} + C$ then $f(3)$ is equal to

Q.27

If $\int \frac{4e^x+6e^{-x}}{9e^x-4e^{-x}}\,dx = Ax + B\log\left(9e^{2x} - 4\right) + C$ then the least integer less than or equal to B-A

Q.28

If $\int \frac{\cos 4x+1}{\cot x-\tan x} = K\cos 4x + C$, then the value of -24k

Q.29

Let $g(x) = \int \frac{1+2\cos x}{(\cos x+2)^2}dx$ and $g(0) = 0$, then the value of $8g(\pi/2)$ is

Q.30

If $\int \left[\left(\frac{x}{e}\right)^x + \left(\frac{e}{x}\right)^x\right]\ln x dx = A\left(\frac{x}{e}\right)^x + B\left(\frac{e}{x}\right)^x + C$, then the value of $A + B$ is

Answers & Solutions

Q.1 Answer:

-2

Solution:

Put $\tan x = t$, **where** $I = 2\int \frac{\sqrt{\cot x}}{\sin 2x}dx = \int t^{-3/2}dt = -2\sqrt{\cot x} + c$

Q.2 Answer:

$\frac{x}{\log x} + c$

Solution:

Put $\log x = t \quad \Rightarrow dx = e^t dt$

$I = \int e^t\left(\frac{1}{t} - \frac{1}{t^2}\right)dt = \frac{e^t}{t} + c = \frac{x}{\log x} + c$

Q.3 Answer:

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$x-\frac{x^2}{2}+\frac{x^3}{3}-\frac{x^4}{4}+\frac{x^5}{5}+c$

Solution:

$\frac{1+x^5}{1+x}=\frac{1-(-x)^5}{1-(-x)}$

(sum of a G.P. whose first term is 1, common ratio '$-x$' and number of terms is 5)

$=1-x+x^2-x^3+x^4$

$\int\frac{1+x^5}{1+x}dx=\int\left(1-x+x^2-x^3+x^4\right)dx=x-\frac{x^2}{2}+\frac{x^3}{3}-\frac{x^4}{4}+\frac{x^5}{5}+c$

Q.4 Answer:

$\frac{(3e)^x}{1+\log 3}+c$

Solution:

$\int 3^x\cdot e^x dx=\int(3.e)^x dx=\frac{(3e)^x}{\log(3e)}+c=\frac{3^x e^x}{1+\log 3}+c$

Q.5 Answer:

$-\frac{1}{2}\sin 2x-\sin x+c$

Solution:

$\frac{\cos 5x+\cos 4x}{1-2\cos 3x}=\frac{\sin 3x(\cos 5x+\cos 4x)}{\sin 3x-\sin 6x}$

$=\frac{\sin 3x\cdot 2\cos\frac{9x}{2}\cos\frac{x}{2}}{-2\cos\frac{9x}{2}\sin\frac{3x}{2}}=\frac{2\sin\frac{3x}{2}\cos\frac{3x}{2}\cos\frac{x}{2}}{-\sin\frac{3x}{2}}$

$=-\left(2\cos\frac{3x}{2}\cos\frac{x}{2}\right)=-(\cos 2x+\cos x)$

$\therefore$ **given integral** $=-\int(\cos 2x+\cos x)dx=-\frac{\sin 2x}{2}-\sin x+c$

Q.6 Answer:

$\frac{x}{(\ln x)^2+1}+c$

Solution:

Put $\ln x=t$

$I=\int e^t\left(\frac{t-1}{t^2+1}\right)^2 dt=\int e^t\left(\frac{1}{t^2+1}-\frac{2t}{(t^2+1)^2}\right)dt=\frac{e^t}{t^2+1}+c=\frac{x}{(\ln x)^2+1}+c$

Q.7 Answer:

$\sin x$

Solution:

Given equation is satisfied if $\cos x dx=d(f(x))\Rightarrow\quad f(x)=\sin x$

Q.8 Answer:

$x+c$

Solution:

Let $\sin x=z\Rightarrow d(\sin x)=dz$

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$\int \frac{dz}{\sqrt{1-z^2}} = \sin^{-1} z + c = \sin^{-1}(\sin x) + c = x + c$

Q.9 Answer:

$e^x \tan\left(\frac{x}{2}\right) + c$

Solution:

$$\int e^x \left(\frac{1+2\sin\left(\frac{x}{2}\right)\cos\left(\frac{x}{2}\right)}{2\cos^2\frac{x}{2}}\right) dx = \int e^x \left(\tan\frac{x}{2} + \frac{1}{2}\sec^2\frac{x}{2}\right) dx = e^x \tan\frac{x}{2} + c$$

Q.10 Answer:

$(x+1)\tan^{-1}\sqrt{x} - \sqrt{x} + C$

Solution:

$$I = \int \tan^{-1}\sqrt{x}dx = x\cdot\tan^{-1}\sqrt{x} - \int \frac{1}{1+x}\cdot\frac{2}{2\sqrt{x}}\cdot x dx = x\cdot\tan^{-1}\sqrt{x} - \frac{1}{2}\int \frac{\sqrt{x}}{1+x}dx$$

Now putting $\sqrt{x} = t$ **i.e.,** $\frac{1}{2\sqrt{x}}dx = dt$

$$I = x\cdot\tan^{-1}\sqrt{x} - \frac{1}{2}\int \frac{t}{1+t^2}2tdt = x\cdot\tan^{-1}\sqrt{x} - \int \frac{t^2+1-1}{1+t^2}dt$$

$$= x\cdot\tan^{-1}\sqrt{x} - t + \tan^{-1} t + C = x\cdot\tan^{-1}\sqrt{x} - \sqrt{x} + \tan^{-1}\sqrt{x} + C$$

Q.11 Answer:

$\frac{1}{2}\left\{\log\frac{\phi(x)}{f(x)}\right\}^2 + k$

Solution:

$$I = \int \log\frac{\phi(x)}{f(x)} d\left\{\log\frac{\phi(x)}{f(x)}\right\} = \frac{1}{2}\left\{\log\frac{\phi(x)}{f(x)}\right\}^2 + k$$

Q.12 Answer:

$\frac{1}{2}\left(x^2 F\left(x^2\right) - \int F\left(x^2\right) dx^2\right)$

Solution:

Put $x^2 = t \Rightarrow x\cdot dx = \frac{dt}{2}$

$$\frac{1}{2}\int t f(t)dt = \frac{1}{2}\left\{t\int f(t)dt - \iint (f(t)dt)dt\right\} = \frac{1}{2}\left\{tF(t) - \int F(t)dt\right\}$$

$$= \frac{1}{2}\left\{x^2 F\left(x^2\right) - \int F\left(x^2\right) d\left(x^2\right)\right\}$$

Q.13 Answer:

$\frac{7}{11}\left(\tan\frac{\theta}{2}\right)^{\frac{11}{7}} + C$

Solution:

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Let $I=\int \frac{(1-\cos\theta)^{2/7}}{(1+\cos\theta)^{9/7}}d\theta$

$,I=\int \frac{\left(2\sin^2\theta/2\right)^{2/7}}{(2\cos^2\theta/2)^{9/2}}d\theta=\frac{1}{2}\int \frac{(\sin\theta/2)^{4/7}}{(\cos\theta/2)^{18/7}}d\theta$

Put $\frac{\theta}{2}=t\therefore\frac{d\theta}{2}=dt$

$\Rightarrow \quad I=\int\frac{(\sin t)^{4/7}}{(\cos t)^{18/7}}dt$

$(\text{Here } m+n=-2)$

$=\int(\tan t)^{4/7}\sec^2 tdt$

Put $\tan t=u\therefore\sec^2 tdt=du$

$$\Rightarrow I=\int u^{4/7}du=\frac{u^{11/7}}{11/7}+c=\frac{7}{11}(\tan t)^{11/7}+C$$

$$=\frac{7}{11}\left(\tan\frac{\theta}{2}\right)^{11/7}+C$$

Q.14 Answer:

-1

Solution:

Substituting $1+e^x=t$, we have

$$\int\frac{dx}{1+e^x}=\int\frac{dt}{t(t-1)}=\int\frac{1}{t-1}dt-\int\frac{dt}{t}$$

$$=\log[t-1]-\log t+C$$

$$=\log e^x-\log\left(1+e^x\right)+C$$

$$=x-\log\left(1+e^x\right)+C$$

Q.15 Answer:

$\left(\sin^{-1}x\right)^3$

Solution:

The integrand is equal to $\frac{2x}{\sqrt{1-x^2}}-\frac{\sqrt{\sin^{-1}x}}{\sqrt{1-x^2}}$.

The first part is of form $\frac{-g'(x)}{\sqrt{g(x)}}$ where $g(x)=\left(1-x^2\right)$

and other part is of the form $\sqrt{h(x)}h'(x), h(x)=\sin^{-1}x$

Hence the required integral $C-2\sqrt{1-x^2}-\frac{2}{3}\left(\sin^{-1}x\right)^{3/2}$.

Thus $f(x)=\left(\sin^{-1}x\right)^3$

Q.16 Answer:

$-\frac{8}{3}\log|1-x|+\frac{2}{3}x+c$

Solution:

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Put $x=\frac{3t-4}{3t+4}=1-\frac{8}{3t+4}, dx=\frac{24}{(3t+4)^2}dt$

$$\begin{aligned}\text{Now } \int f(x)dx &= \int f\left(\frac{3t-4}{3t+4}\right)\frac{24}{(3t+4)^2}dt \\ &= \int \frac{t+2}{(3t+4)^2}(24)dt \\ &= 8\int \frac{3t+4+2}{(3t+4)^2}dt \\ &= 8\int \left[\frac{1}{3t+4}+\frac{2}{(3t+4)^2}\right]dt \\ &= \frac{8}{3}\log|3t+4|-\frac{2}{3}\frac{8}{3t+4}+C_1 \\ &= \frac{8}{3}\log\left|\frac{8}{1-x}\right|-\frac{2}{3}(1-x)+C_1 \\ &= -\frac{8}{3}\log|1-x|+\frac{2}{3}x+C\end{aligned}$$

Q.17 Answer:

$C-\frac{1}{2x^2}\log(x\sqrt{e})$

Solution:

$F(x)=\int\frac{\log x}{x^3}dx=\log x\int\frac{1}{x^3}dx-$
$\int\frac{1}{x}\left(\int\frac{1}{x^3}dx\right)dx$ (integration by parts)

$$\begin{aligned}&= -\frac{1}{2x^2}\log x+\frac{1}{2}\int\frac{1}{x^3}dx+\text{ Constant} \\ &= -\frac{1}{2x^2}\log x-\frac{1}{4x^2}+C \\ &= -\frac{1}{2x^2}\left[\log x+\frac{1}{2}\log e\right]+C \\ &= -\frac{1}{2x^2}\log x\sqrt{e}+C\end{aligned}$$

Q.18 Answer:

$\frac{1}{8}\log\left|\frac{\sin x-1}{\sin x+1}\right|-\frac{1}{4\sqrt{2}}\log\left|\frac{\sqrt{2}\sin x-1}{\sqrt{2}\sin x+1}\right|+C$

Solution:

The given integrand can be written as $\frac{1}{4\cos x\cos 2x}=\frac{\cos x}{4(1-\sin^2 x)(1-2\sin^2 x)}$,
hence $\int\frac{\sin x}{\sin 4x}\,dx=\frac{1}{4}\int\frac{dt}{(1-t^2)(1-2t^2)}\quad(t=\sin x)$
$=\frac{1}{4}\int\left[\frac{1}{t^2-1}-\frac{1}{t^2-1/2}\right]dt$

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$= \frac{1}{4}\left[\frac{1}{2}\log\left|\frac{t-1}{t+1}\right| - \frac{1}{\sqrt{2}}\log\left|\frac{t-1/\sqrt{2}}{t+1/\sqrt{2}}\right|\right] + C$
$= \frac{1}{8}\log\left|\frac{\sin x-1}{\sin x+1}\right| - \frac{1}{4\sqrt{2}}\log\left|\frac{\sqrt{2}\sin x-1}{\sqrt{2}\sin x+1}\right| + C.$

Q.19 Answer:

$\frac{1}{2}\log\left(\frac{e^{2x}+e^{x}+1}{e^{2x}-e^{x}+1}\right) + \mathrm{C}$

Solution:

$I - J = \int \frac{e^x}{e^{4x}+e^{2x}+1}dx - \int \frac{e^{3x}}{e^{4x}+e^{2x}+1}dx$
$= \int \frac{e^x\left(1-e^{2x}\right)}{e^{4x}+e^{2x}+1}dx$
Put $e^x = t, e^x dx = dt$
$I - J = \int \frac{\left(1-t^2\right)}{t^4+t^2+1}dt = \int \frac{\frac{1}{t^2}-1}{t^2+\frac{1}{t^2}+1}dt$
$1 - \frac{1}{t^2}$
$u = t + \frac{1}{t}$
$= \frac{1}{2}\log\left|\frac{1+u}{1-u}\right| + C$
$= \frac{1}{2}\log\left|\frac{e^x+e^{-x}+1}{e^x+e^{-x}-1}\right| + C$
$= \frac{1}{2}\log\left|\frac{e^{2x}+e^{x}+1}{e^{2x}-e^{x}+1}\right| + C.$
$= \frac{1}{2}\log\left|\frac{1+u}{1-u}\right| + C$
$= \frac{1}{2}\log\left|\frac{e^x+e^{-x}+1}{e^x+e^{-x}-1}\right| + C$
$= \frac{1}{2}\log\left|\frac{e^{2x}+e^{x}+1}{e^{2x}-e^{x}+1}\right| + C.$

Q.20 Answer:

$-e^{\tan x}\cos x + C$

Solution:

$I = \int e^{\tan x}(\sin x - \sec x)dx$
$= \int \sin x e^{\tan x}dx - \int \sec x e^{\tan x}dx$
$= -e^{\tan x}\cos x + \int \cos x e^{\tan x}\sec^2 x dx - \int \sec x e^{\tan x}dx$
$= -\cos x e^{\tan x} + C$

Q.21 Answer:

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Solution:

Putting tan $x = t$, the given integral can be written as $\int t^2\left(1+t^2\right)\mathrm{d}t = \frac{t^3}{3} + \frac{t^5}{5} + C = \frac{\tan^5 x}{5} + \frac{\tan^3 x}{3} + C$ **which is a polynomial of degree 5 in** $\tan x$.

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Q.22 Answer:

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Solution:

$F(x) = \int f(x)\mathrm{d}x + C = x + 3^x + C$ **but** $F(2) = 7$ **so** $7 = 2 + 9 + C$
$\Rightarrow C = -4.$
Hence $F(x) = x + 3^x - 4.$
Now $F(x) = 0$
$\Rightarrow x = 1.$

Q.23 Answer:

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Solution:

Let the two antiderivates be $e^x + C_1$ **and** $e^x + C_2$
At $x = 1, C_1 - C_2 = 2$
At $x = 100$ **it is** $e^{100} + C_1 - e^{100} - C_2 = C_1 - C_2 = 2$

Q.24 Answer:

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Solution:

Let $I = \int \frac{5\tan x}{\tan x - 2}dx = \int \frac{5\sin x}{\sin x - 2\cos x}dx$
Let $5\sin x = A(\sin x - 2\cos x) + B(\cos x + 2\sin x)$**. Equating coefficients of** $\sin x$ **and** $\cos x$**, we get** $5 = A + 2B$ **and** $0 = -2A + B \Rightarrow A = 1, B = 2$

$\therefore I = \int \frac{\sin x - 2\cos x + 2(\cos x + 2\sin x)}{\sin x - 2\cos x}dx$
$= \int \left[1 + \frac{2(\cos x + 2\sin x)}{\sin x - 2\cos x}\right]dx$
$= x + 2\,\mathrm{In}\,|\sin x - 2\cos x| + k$
$\therefore a = 2$

Q.25 Answer:

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Solution:

$f(x) = \int x^{\sin x}(1 + x\cos x \cdot \ln x + \sin x)dx$
if $F(x) = x^{\sin x} = e^{\sin x \ln x}$
$\therefore f(x) = \int (F(x) + xF'(x)) = xF(x) + C$

$$f(x) = x \cdot x^{\sin x} + C$$
$$f\left(\frac{\pi}{2}\right) = \frac{\pi}{2} \cdot \frac{\pi}{2} + C \quad \Rightarrow \quad C = 0$$
$$\therefore f(x) = x(x)^{\sin x}; f(\pi) = \pi(\pi)^0 = \pi$$

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Q.26 Answer:

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Solution:

Differentiating both the sides w.r.t. x, we get

$\frac{f(x)}{x^2-x+1} = \frac{3}{2}\frac{2x-1}{x^2-x+1} + \frac{1}{\sqrt{3}}\frac{1}{1+(2x-1)^{2/3}}\left(\frac{2}{\sqrt{3}}\right)$

$= \frac{3}{2}\frac{2x-1}{x^2-x+1} + \frac{1}{2(x^2-x+1)}$

$\Rightarrow f(x) = 3x - 1$

Q.27 Answer:

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Solution:

Let $I = \int \frac{4e^x+6e^{-x}}{9e^x-4e^{-x}}dx$

$= \int \frac{4e^{2x}+6}{9e^{2x}-4}dx$

Put $e^{2x} = t$**, so that**

$I = \int \frac{4t+6}{9t-4}\cdot\frac{1}{2t}dt$

$= \int \left[\frac{-3}{4}\frac{1}{t} + \frac{35}{4}\frac{1}{9t-4}\right]dt$

$= \frac{-3}{4}\ln|t| + \frac{35}{36}\ln|9t-4| + C$

$= \frac{-3}{2}x + \frac{35}{36}\ln\left(9e^{2x}-4\right) + C$

$\therefore A = \frac{-3}{2}, B = \frac{35}{36}$

Alternatively differentiate both the sides w.r.t. x**, we get**

$\frac{4e^x+6e^{-x}}{9e^x-4e^{-x}} = A + \frac{18Be^{2x}}{9e^{2x}-4}$

$\Rightarrow 4e^x + 6e^{-x} = A\left(9e^x - 4^{e-x}\right) + 18Be^x$

Equating the coefficients of e^x **and** e^{-x} **We get** $4 = 9A + 18B$ **and** $6 = -4A$

$\Rightarrow A = \frac{-3}{2}, B = \frac{35}{36}$

Q.28 Answer:

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Solution:

$\int \frac{\cos 4x+1}{\cot x-\tan x}\,\mathrm{d}x$

$= \int \frac{2\cos^2 2x}{\cos^2 x-\sin^2 x}\cdot \sin x\cos x\,\mathrm{d}x$

$= \int \cos 2x \sin 2x\,\mathrm{d}x = \frac{1}{2}\int \sin 4x\,\mathrm{d}x$

$= -\frac{1}{8}\cos 4x + C$

$= -1/8.$

Hence $K = -1/8$

Q.29 Answer:

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Solution:

$g(x)=\int \frac{\cos x(\cos x+2)+\sin^2 x}{(\cos x+2)^2}dx$

$=\int \underbrace{\cos x}_{\text{II}}\cdot \underbrace{\frac{1}{(\cos x+2)_I}}_{\text{I}}dx+\int \frac{\sin^2 x}{\cos x+2}dx$

$=\frac{1}{\cos x+2}\cdot \sin x-\int \frac{\sin^2 x}{(\cos x+2)^2}dx+\int \frac{\sin^2 x}{(\cos x+2)^2}dx$

$\therefore g(x)=\frac{\sin x}{\cos x+2}+C$

$\therefore g(0)=0\Rightarrow C=0$

$\therefore g(x)=\frac{\sin x}{\cos x+2}\Rightarrow g\left(\frac{\pi}{2}\right)=\frac{1}{2}$

Q.30 Answer:

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Solution:

$\int \left[\left(\frac{x}{e}\right)^x+\left(\frac{e}{x}\right)^x\right]\ln x dx$

put $\left(\frac{x}{e}\right)^x=t$

or $x\ln\left(\frac{x}{e}\right)=\ln t$

$\therefore \left(x\cdot\frac{1}{x/e}\cdot\frac{1}{e}+\ln\left(\frac{x}{e}\right)\right)dx=\frac{1}{t}dt$

$\therefore (1+\ln x-\ln e)dx=\frac{1}{t}dt$

$\therefore (\ln e+\ln x-\ln e)dx=\frac{1}{t}dt$

$\therefore (\ln x)dx=\frac{1}{t}dt$

or $I=\int\left(t+\frac{1}{t}\right)\frac{1}{t}dt=\int 1\cdot dt+\int\frac{1}{t^2}dt$

$=t-\frac{1}{t}+C$

or $I=\left(\frac{x}{e}\right)^x-\left(\frac{e}{x}\right)^x+C$

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